CHAPTER

2 TESTS BASED ON SAMPLING DISTRIBUTIONS I





W.S. Gossett

W Gosset (1876-1937), born in England studied Chemistry and Mathematics at New College , Oxford. Upon graduating in 1899, he joined a brewery in Ireland. Gosset applied his statistical knowledge both in the brewery and on the farm to the selection of the best varieties of Barley. Gosset acquired that knowledge by study, by trial and error, and by spending two terms in 1906–1907 in the biometrical laboratory of Karl Pearson. Gosset and Pearson had a good relationship. Pearson helped Gosset with the mathematics of his research papers. The brewery where he was employed allowed publishing his work under a pseudonym ("Student"). Thus, his most noteworthy achievement is now called Student's *t*, rather than Gosset's, *t*-distribution.

LEARNING OBJECTIVES

B1BNMF

The student will be able to

- understand the purpose for using *t*-test and chi-square test .
- understand procedures for tests of hypotheses based on small samples.
- solve problems to test the hypotheses concerning mean(s) using *t*-distribution.
- solve problems to test the hypothesis whether the population has a particular variance using chi-square test.
- solve problems to test the hypotheses relating to independence of attributes and goodness of fit using chi-square test.

Introduction

In the earlier chapter, we have discussed various problems related to tests of significance based on large samples by applying the standard normal distribution. However, if the sample size is small ($n < 30$) the sampling distributions of test statistics are far from normal and the procedures discussed in Chapter-1 cannot be applied, except the general procedure (Section 1.8). But in this case, there exists a probability distribution called *t*-distribution which may be used instead of standard normal distribution to study the problems based on small samples.

12th_Statistics_EM_Unit_2.indd 37 1/28/2020 7:09:34 PM